

# Chapter 1

## BCR (Benefit-Cost Ratio) Cheat Sheet

Quick reference guide for all BCR perspectives calculated by CTCC.

### Important Note on Revenue

Revenue (rate-based revenue requirement) is a **transfer** from ratepayers to the utility/TSP, not a net social benefit. System/societal BCRs exclude revenue from total benefits. Revenue appears only in stakeholder BCRs where it's a benefit to utilities and a cost to ratepayers.

### Cost categories

**Operational** = O&M + insurance only. **Energy/Emissions** = line losses + emissions + residual exceedance. Residual exceedance is a system/societal cost only (included in total costs and in Energy/Emissions; not included in utility costs or ratepayer costs).

## 1.1 BCR Metrics (order matches Benefit-Cost Ratio Metrics tables)

Order below matches the per-scenario and all-scenarios BCR tables in `results_tables.tex` (updated by `papers/update_results_tables.py`).

### 1.1.1 `bcr_system`

**Equation:** Total Benefits (haircut) / Total Costs

**Perspective:** Full societal/system-wide analysis

**Includes:**

- **Benefits:** Congestion reduction + Curtailment reduction (with conservative haircuts; excludes revenue transfer)
- **Costs:** Capital + Operational (O&M + insurance only) + Energy/Emissions (line losses + emissions + residual exceedance) + Risk + Delay (everything)

**Use Case:** Complete societal cost-benefit analysis including all externalities and risks.

### 1.1.2 `bcr_utility`

**Equation:**  $\text{Revenue} / (\text{Capital Costs} + \text{Delay Costs} + \text{Operational Costs})$ . Benefits: Revenue only. Costs: Capital + Delay + Operational (O&M + insurance only). Excludes: Line losses, Emissions, Residual exceedance, Risk.

**Key Insight:** Utilities don't receive congestion/curtailment benefits directly; they only get rate base recovery.

### 1.1.3 `bcr_ratepayer`

**Equation:**  $(\text{Congestion Benefits} + \text{Curtailment Benefits}) / (\text{Revenue} + \text{Line Loss Costs})$ . Benefits: Congestion + Curtailment. Costs: Revenue + Line losses. Excludes: Capital/O&M as separate line items (recovered via Revenue).

**Key Insight:** Ratepayers receive congestion/curtailment relief but pay revenue/rate base and line losses through rates.

### 1.1.4 `bcr_capital`

**Equation:**  $\text{Total Benefits (haircut)} / \text{Capital Costs}$ . Costs: Build + ROW + environmental only.

### 1.1.5 `bcr_capital_and_delay`

**Equation:**  $\text{Total Benefits (haircut)} / (\text{Capital Costs} + \text{Delay Costs})$ . Costs: Capital + Delay (construction delay + congestion delay + curtailment delay).

### 1.1.6 `bcr_primary`

**Equation:**  $\text{Primary Benefits} / \text{Primary Costs}$  (customizable via flags). Benefits: Congestion (optional) + Curtailment (optional). Costs: Capital + Delay (always) + O&M, Insurance, Wildfire, Outage, Line Losses, Emissions (optional). If no flags are set, equals `bcr_system`.

### 1.1.7 `bcr_excluding_wildfire_risk_and_outage_risk`

**Equation:**  $\text{Total Benefits (haircut)} / (\text{Total Costs} - \text{Wildfire Risk} - \text{Outage Risk})$

**Perspective:** Societal perspective excluding both wildfire and outage risk costs.

**Includes:** Benefits: Congestion + Curtailment (excludes revenue). Costs: Capital + Operational + Energy/Emissions + Delay (excludes wildfire + wildfire liability + outage).

**Use Case:** Analysis when both wildfire and outage risk costs may be insured, managed separately, or contextually inappropriate.

### 1.1.8 `bcr_excluding_emissions`

**Equation:**  $\text{Total Benefits (haircut)} / (\text{Total Costs} - \text{Emissions Costs})$ . Costs: Capital + Operational + Risk + Delay + Line Losses (excludes emissions only).

### 1.1.9 `bcr_excluding_linelosses`

**Equation:** Total Benefits (haircut) / (Total Costs - Line Loss Costs). Costs: Capital + Operational + Risk + Delay + Emissions (excludes line losses only).

### 1.1.10 `bcr_excluding_emissions_and_linelosses`

**Equation:** Total Benefits (haircut) / (Total Costs - Emissions Costs - Line Loss Costs). Costs: Capital + Operational + Risk + Delay (excludes emissions + line losses).

### 1.1.11 `bcr_excluding_emissions_and_wildfire_risk_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay + Line Losses.

### 1.1.12 `bcr_excluding_linelosses_and_wildfire_risk_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Line Losses - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay + Emissions.

### 1.1.13 `bcr_excluding_emissions_and_linelosses_and_wildfire_risk_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Line Losses - Wildfire Risk - Outage Risk). Costs: Capital + Operational + Delay only.

### 1.1.14 `bcr_excluding_wildfire_risk`

**Equation:** Total Benefits (haircut) / (Total Costs - Wildfire Risk Costs). Costs: Capital + Operational + Energy/Emissions + Delay + Outage Risk (excludes wildfire + wildfire liability).

### 1.1.15 `bcr_excluding_emissions_and_wildfire_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Wildfire Risk). Costs: Capital + Operational + Delay + Line Losses + Outage Risk.

### 1.1.16 `bcr_excluding_linelosses_and_wildfire_risk`

**Equation:** Total Benefits / (Total Costs - Line Loss Costs - Wildfire Risk Costs). Costs: Capital + Operational + Delay + Emissions + Outage Risk.

### 1.1.17 `bcr_excluding_emissions_and_linelosses_and_wildfire_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Line Losses - Wildfire Risk). Costs: Capital + Operational + Delay + Outage Risk only.

### 1.1.18 `bcr_excluding_outage_risk`

**Equation:** Total Benefits (haircut) / (Total Costs - Outage Risk Costs). Costs: Capital + Operational + Energy/Emissions + Delay + Wildfire Risk (excludes outage).

### 1.1.19 `bcr_excluding_emissions_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Outage Risk). Costs: Capital + Operational + Delay + Line Losses + Wildfire Risk.

### 1.1.20 `bcr_excluding_linelosses_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Line Loss Costs - Outage Risk Costs). Costs: Capital + Operational + Delay + Emissions + Wildfire Risk.

### 1.1.21 `bcr_excluding_emissions_and_linelosses_and_outage_risk`

**Equation:** Total Benefits / (Total Costs - Emissions - Line Losses - Outage Risk). Costs: Capital + Operational + Delay + Wildfire Risk only.

## 1.2 Net Benefits

All BCRs have corresponding net benefit calculations (Present Value unless noted).

### 1.2.1 System/Societal Net Benefits

- `net_benefit_pv` = Total Benefits (haircut) - Total Costs
- `net_benefit_nominal` = Total Benefits (nominal) - Total Costs (nominal), undiscounted
- `net_benefit_primary_pv` = Primary Benefits - Primary Costs (customizable)
- `net_benefit_excluding_wildfire_risk_and_outage_risk_pv` = Total Benefits - (Total Costs - Wildfire Risk - Outage Risk)
- `net_benefit_excluding_wildfire_risk_pv`, `net_benefit_excluding_outage_risk_pv`
- `net_benefit_excluding_emissions_and_linelosses_pv`
- `net_benefit_excluding_emissions_and_linelosses_and_wildfire_risk_and_outage_risk_pv`
- `net_benefit_capital_only_pv`, `net_benefit_capital_and_delay_pv`

### 1.2.2 Stakeholder Net Benefits

- `net_benefit_utility_pv` = Revenue - (Capital + Delay + Operational Costs)
- `net_benefit_ratepayer_pv` = (Congestion + Curtailment Benefits) - (Revenue + Line Loss Costs)

### 1.2.3 Partial Exclusion Net Benefits (16 Combinations)

Systematic exploration of all combinations excluding emissions, line losses, wildfire risk, and outage risk. All use societal perspective (benefits: congestion + curtailment; excludes revenue transfer). Positive = benefits exceed costs; Negative = costs exceed benefits.

| WF | Out | Em | LL | Net Benefit Variable                              | Equation                                |
|----|-----|----|----|---------------------------------------------------|-----------------------------------------|
| N  | N   | N  | N  | <code>net.benefit_pv</code>                       | Benefits - Total Costs                  |
| Y  | N   | N  | N  | <code>...excl. wildfire_risk_pv</code>            | Benefits - (Costs - WF)                 |
| N  | Y   | N  | N  | <code>...excl. outage_risk_pv</code>              | Benefits - (Costs - Outage)             |
| Y  | Y   | N  | N  | <code>...excl. wildfire_and_outage_pv</code>      | Benefits - (Costs - WF - Out)           |
| N  | N   | Y  | N  | <code>...excl. emissions_pv</code>                | Benefits - (Costs - Em); keeps LL       |
| Y  | N   | Y  | N  | <code>...excl. emissions_and_wildfire_pv</code>   | Benefits - (Costs - Em - WF)            |
| N  | Y   | Y  | N  | <code>...excl. emissions_and_outage_pv</code>     | Benefits - (Costs - Em - Out)           |
| Y  | Y   | Y  | N  | <code>...excl. em_wf_outage_pv</code>             | Benefits - (Costs - Em - WF - Out)      |
| N  | N   | N  | Y  | <code>...excl. linelosses_pv</code>               | Benefits - (Costs - LL); keeps Em       |
| Y  | N   | N  | Y  | <code>...excl. LL_wildfire_pv</code>              | Benefits - (Costs - LL - WF)            |
| N  | Y   | N  | Y  | <code>...excl. LL_outage_pv</code>                | Benefits - (Costs - LL - Out)           |
| Y  | Y   | N  | Y  | <code>...excl. LL_wf_outage_pv</code>             | Benefits - (Costs - LL - WF - Out)      |
| N  | N   | Y  | Y  | <code>...excl. emissions_and_linelosses_pv</code> | Benefits - (Costs - Em - LL)            |
| Y  | N   | Y  | Y  | <code>...excl. em_LL_wildfire_pv</code>           | Benefits - (Costs - Em - LL - WF)       |
| N  | Y   | Y  | Y  | <code>...excl. em_LL_outage_pv</code>             | Benefits - (Costs - Em - LL - Out)      |
| Y  | Y   | Y  | Y  | <code>...excl. em_LL_wf_outage_pv</code>          | Benefits - (Costs - Em - LL - WF - Out) |

Column key: WF = wildfire risk, Out = outage risk, Em = emissions, LL = line losses. “...” = `net_benefit_excluding_`.

## 1.3 Quick Decision Guide

| Question                                                  | Use This BCR                                                                           |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------|
| Should a utility/TSP build this?                          | <code>bcr_utility</code>                                                               |
| Do ratepayers benefit?                                    | <code>bcr_ratepayer</code>                                                             |
| Is it good for society overall?                           | <code>bcr_system</code> or <code>bcr_excluding_wildfire_risk_and_outage_risk</code>    |
| Is the capital investment attractive?                     | <code>bcr_capital</code> or <code>bcr_capital_and_delay</code>                         |
| Ignore wildfire risk (keep outage)?                       | <code>bcr_excluding_wildfire_risk</code>                                               |
| Ignore outage risk (keep wildfire)?                       | <code>bcr_excluding_outage_risk</code>                                                 |
| Ignore both wildfire and outage risk?                     | <code>bcr_excluding_wildfire_risk_and_outage_risk</code>                               |
| Ignore emissions only?                                    | <code>bcr_excluding_emissions</code>                                                   |
| Ignore line losses only?                                  | <code>bcr_excluding_line losses</code>                                                 |
| Ignore both emissions and line losses?                    | <code>bcr_excluding_emissions_and_line losses</code>                                   |
| Ignore emissions and both risks?                          | <code>bcr_excluding_emissions_and_wildfire_risk_and_outage_risk</code>                 |
| Ignore line losses and both risks?                        | <code>bcr_excluding_line losses_and_wildfire_risk_and_outage_risk</code>               |
| Ignore emissions and wildfire risk (keep outage)?         | <code>bcr_excluding_emissions_and_wildfire_risk</code>                                 |
| Ignore emissions and outage risk (keep wildfire)?         | <code>bcr_excluding_emissions_and_outage_risk</code>                                   |
| Core project economics only (no externalities, no risks)? | <code>bcr_excluding_emissions_and_line losses_and_wildfire_risk_and_outage_risk</code> |

## 1.4 Important Notes

1. **Haircut Benefits:** Most BCRs use conservative “haircut” benefits (saturation factors applied to congestion/curtailment).
2. **Present Value:** All values are in present value (PV) terms, discounted to base year.
3. **Context:** Wildfire costs assume California-style catastrophic events. For lower wildfire risk regions, `bcr_excluding_wildfire_risk` may be more appropriate. Outage risk is more universal. Line losses are socialized through rates; emissions are externalities.
4. **Separating Wildfire and Outage Risk:** Use `bcr_excluding_wildfire_risk` for Virginia-style scenarios; `bcr_excluding_wildfire_risk_and_outage_risk` when both should be excluded; `bcr_excluding_outage_risk` when only outage risk should be excluded.

## 1.5 Example: Scenario 1 (Rural Overhead AC 657MW)

- `bcr_utility` = 0.36 → Utility loses money.

- `bcr_ratepayer` = 0.22  $\rightarrow$  Ratepayers lose money.
- `bcr_system` = 0.02  $\rightarrow$  Very poor societal return (wildfire risk dominates).
- `bcr_excluding_wildfire_risk_and_outage_risk` = 0.24  $\rightarrow$  Better but still negative.

The separation of wildfire and outage risk allows for more nuanced regional analysis.